

Math 372: Homework 09

Due Tuesday, March 31

Instructions: Do not write your solutions on this sheet—you don't have enough space to show your work. So use separate sheets of paper. For all problems you need to show your work.

Problem 1 (Exercise 5.1.1). A service station has both self-service and full-service islands. On each island, there is a single regular unleaded pump with two hoses. Let X denote the number of hoses being used on the self-service island at a particular time, and let Y denote the number of hoses on the full-service island in use at that time. The joint pmf of X and Y is described by the following table:

$p(x, y)$	$y = 0$	$y = 1$	$y = 2$
$x = 0$	0.10	0.08	0.06
$x = 1$	0.04	0.20	0.14
$x = 2$	0.02	0.06	0.30

- (a) What is $\mathbb{P}[X = 1 \text{ and } Y = 1]$?
- (b) Compute $\mathbb{P}[X \leq 1 \text{ and } Y \leq 1]$?
- (c) Using words, describe the event $[X \neq 0 \text{ and } Y \neq 0]$, and then compute its probability.
- (d) Compute the marginal pmf of X and Y . Using $p_X(x)$, what is $\mathbb{P}[X \leq 1]$?
- (e) Are X and Y independent random variables?

Problem 2 (Exercise 5.1.3). A grocery store has both an express checkout lane and a superexpress checkout lane. Let X_1 denote the number of customers in line at the express checkout at a particular time of day, and let X_2 denote the number of customers in line at the superexpress checkout at the same time. Suppose the joint pmf of X_1 and X_2 is given by the following table:

	$x_2 = 0$	$x_2 = 1$	$x_2 = 2$	$x_2 = 3$	$x_2 = 4$
$x_1 = 0$	0.08	0.06	0.05	0.00	0.00
$x_1 = 1$	0.07	0.15	0.04	0.03	0.01
$x_1 = 2$	0.04	0.05	0.10	0.04	0.05
$x_1 = 3$	0.00	0.04	0.06	0.07	0.06

- (a) What $\mathbb{P}[X_1 = 1, X_2 = 1]$, that is, the probability that there is exactly one customer in each line?
- (b) What is $\mathbb{P}[X_1 = X_2]$, that is, the probability that the number of customers in the two lines is identical?
- (c) Let A denote the event that there are at least two more customers in one line than in the other? Express A in terms of X_1 and X_2 and calculate the probability of this event.
- (d) What is the probability that the total number of customers in the two lines is exactly four? At least four?

Problem 3 (Exercise 5.1.4). Return to the situation described in Theorem 2.

- (a) Determine the marginal pmf of X_2 , and then calculate the expected number of customers in line at the express checkout.
- (b) Determine the marginal pmf of X_1 .
- (c) By inspection of the probabilities $\mathbb{P}[X_2 = 4]$, $\mathbb{P}[X_1 = 0]$, and $\mathbb{P}[X_2 = 4, X_1 = 0]$, are X_1 and X_2 independent random variables? Explain.

Problem 4 (Exercise 4.3.46). The Rockwell hardness of a metal is determined by impressing a hardened point into the surface of the metal and then measuring the depth of penetration of the point. Suppose the Rockwell hardness of a particular alloy is normally distributed with mean 70 and standard deviation 3.

- (a) If a specimen is acceptable only if its hardness is between 67 and 75, what is the probability that a randomly chosen specimen has an acceptable hardness?
- (b) If the acceptable range of hardness is $(70 - c, 70 + c)$, for what value of c would 95% of all specimens have acceptable hardness?
- (c) If the acceptable range is as in part (a) and the hardness of each of ten randomly selected specimens is independently determined, what is the expected number of acceptable specimens among the ten?
- (d) What is the probability that at most eight of ten independently selected specimens have a hardness of less than 73.84? [Hint: If Y is the number among the ten specimens with hardness less than 73.84, then Y is a binomial variable; what is p ?

Problem 5. Annie and Alvie have agreed to meet for lunch between noon and 1:00pm. Denote Annie's arrival time by X and Alvie's by Y . Suppose that X and Y are independent with pdfs

$$f_X(x) = \begin{cases} 3x^2 & : 0 \leq x \leq 1 \\ 0 & : \text{otherwise} \end{cases} \quad \text{and} \quad f_Y(y) = \begin{cases} 2y & : 0 \leq y \leq 1 \\ 0 & : \text{otherwise} \end{cases}$$

What is the expected amount of time that the one who arrives first must wait for the other person. *Hint:* $h(X, Y) = |X - Y|$

Problem 6. Let (X, Y) be uniformly distributed on the unit square $0 < x < 1$, $0 < y < 1$. Find the following probabilities: [*hint: sketch the regions in the unit square*]

- (a) $\mathbb{P}[X < Y]$
- (b) $\mathbb{P}[X + Y < 1]$
- (c) $\mathbb{P}[\min(X, Y) > \frac{3}{4}]$
- (d) $\mathbb{P}[XY \leq \frac{1}{2}]$